

Evaluation of Privacy-Preserving Techniques for Edge Computing

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Privacy Concerns

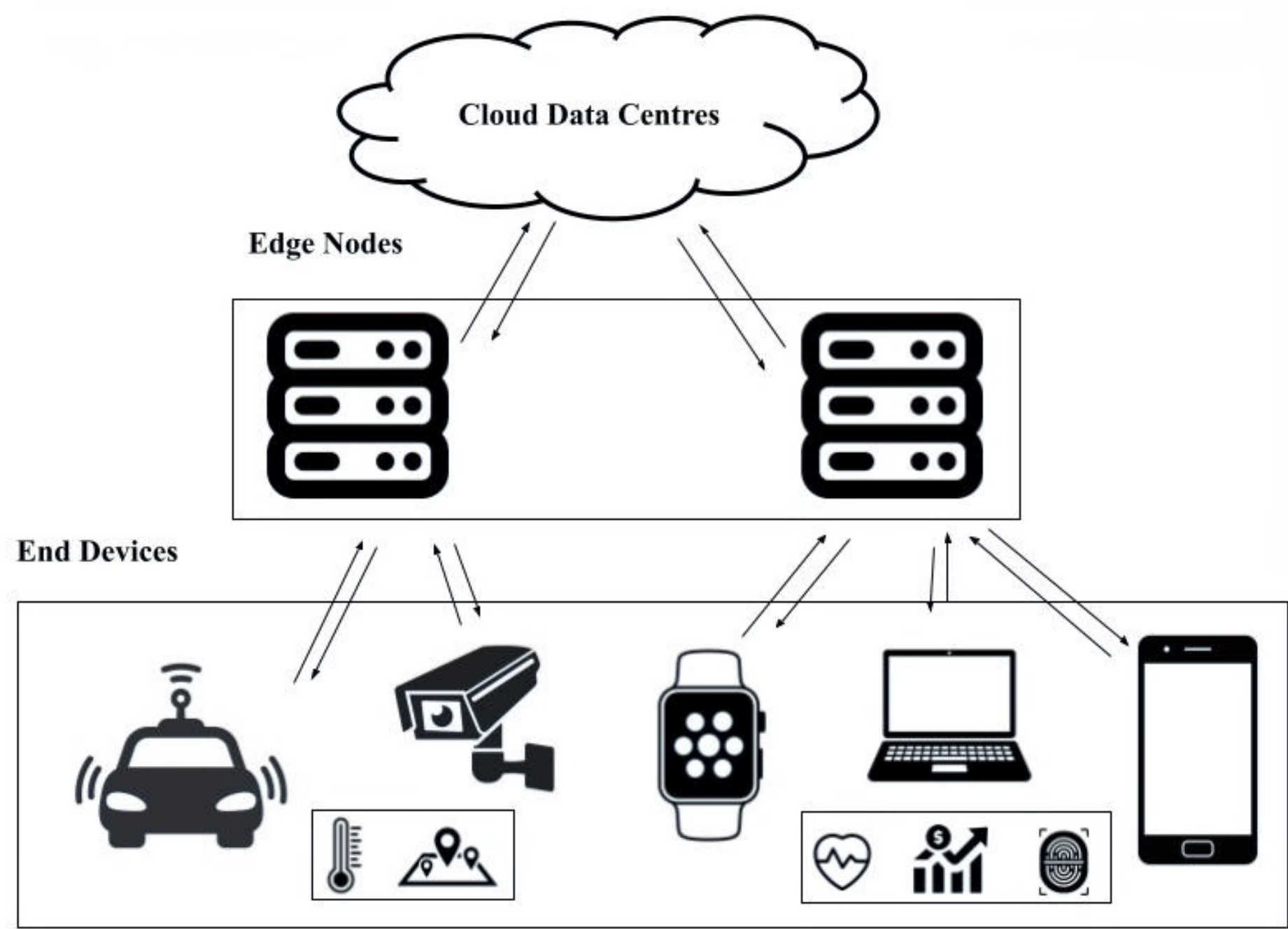
A primary concern in today's digital landscape is the risk of personal data breaches during transmission, storage, and analysis, which can lead to unauthorized access and misuse [1].



Essential Privacy Aspects

Privacy at the Edge

- Edge computing processes large data volumes from IoT systems, connected vehicles, and smart communities locally on edge servers.
- Local data processing reduces the volume of sensitive data sent to and stored in the cloud, thereby enhancing data privacy.
- Preserving edge privacy ensures user data control, enhances utility by fostering trust and secure usage.



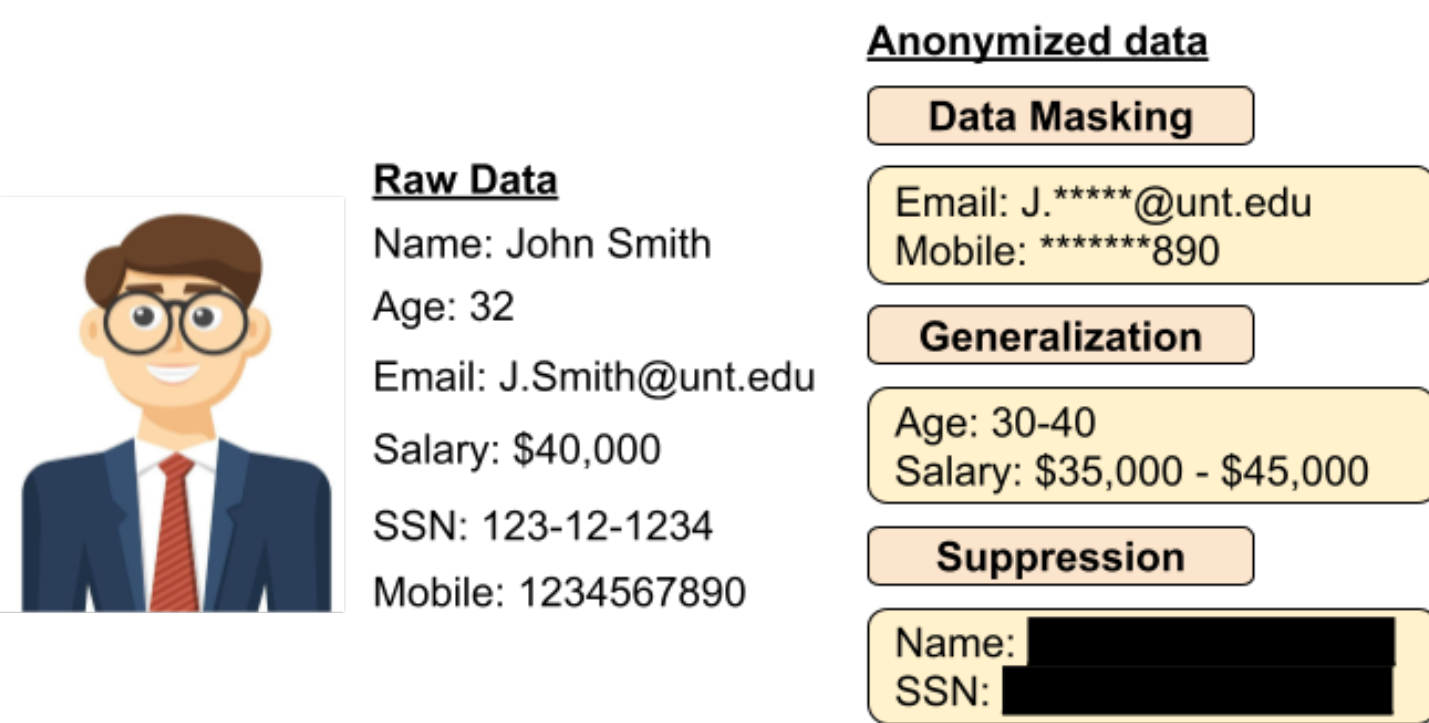
Edge-to-cloud transmission of data from IoT devices

Challenges

- Scalability:** Increasing IoT devices strain the limited computational and storage capacities of edge devices, making it difficult to implement robust privacy mechanisms.
- Implementation Complexity:** The decentralized and diverse edge devices lack standardized privacy frameworks, complicating implementation across platforms.
- Latency Concerns:** Privacy techniques like encryption and anonymization introduce processing delays, increasing latency in real-time applications, a critical issue in edge computing.
- Balancing Data Utility and Privacy:** Privacy-preserving techniques, such as encryption and anonymization, can reduce the usefulness of data, making it challenging to maintain both data privacy and utility in edge computing.

Privacy Techniques

Traditional Data Anonymization – In tradition, data anonymization is a common and effective method for safeguarding privacy when publishing data [2]. Data anonymization techniques involve data masking, pseudonymization, generalization, data swapping, and suppression.



An Example of Traditional Data Anonymization

K-Anonymity is a technique to prevent link attacks by generalizing and/or suppressing certain elements. This ensures that no individual is uniquely identifiable within a group of size k. K is a parameter that represents the level of anonymity.

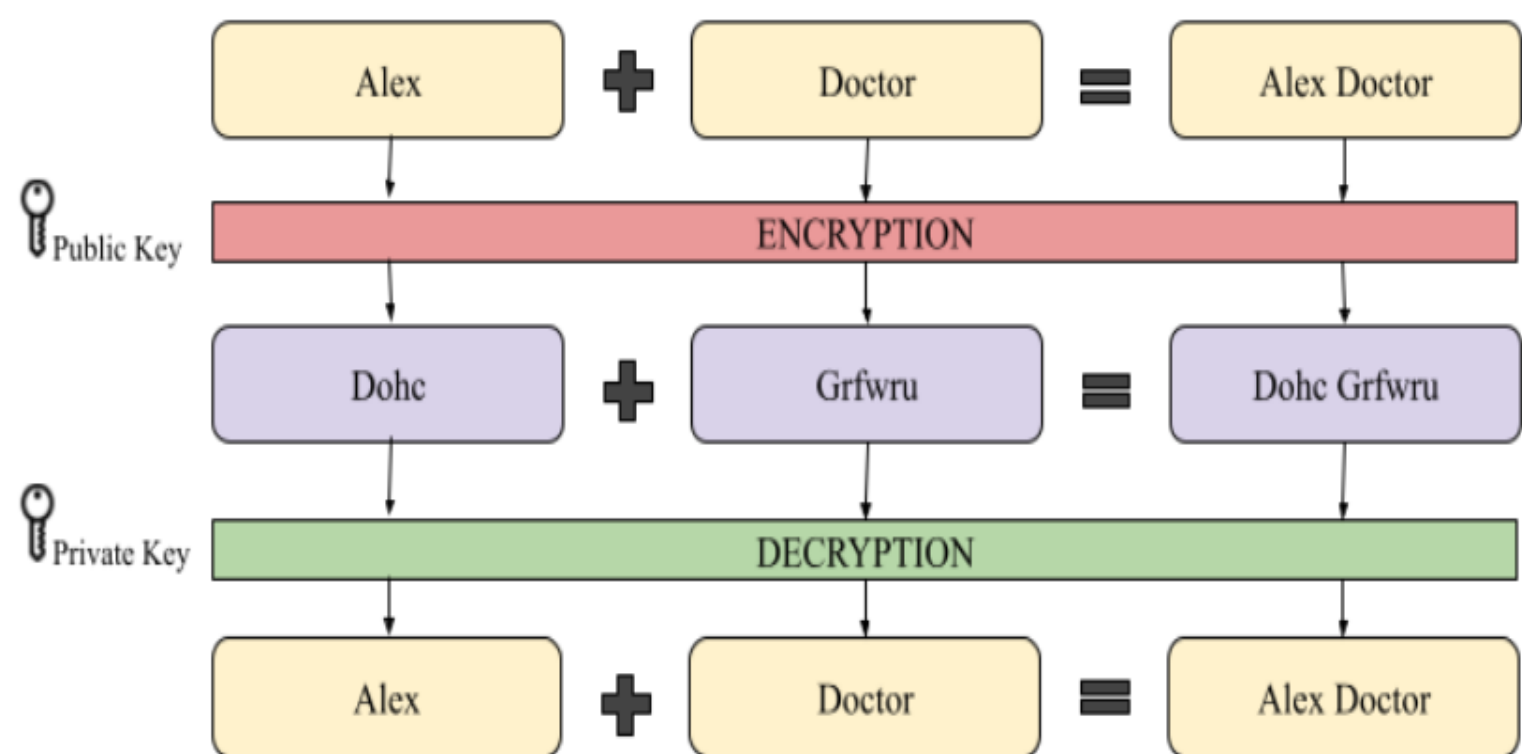
Name	Age	Gender	Zip Code	Disease
Dana	24	F	45011	Diabetes
John	30	M	45012	None
Jack	28	M	45013	Asthma
Alex	26	M	45014	Hypertension
Chloe	27	F	45015	None
Fiona	29	F	45016	Diabetes

Original Dataset

Name	Age	Gender	Zip Code	Disease
Suppressed	21-25	F	450**	Diabetes
Suppressed	25-30	M	450**	None
Suppressed	25-30	M	450**	Asthma
Suppressed	25-30	M	450**	Hypertension
Suppressed	25-30	F	450**	None
Suppressed	25-30	F	450**	Diabetes

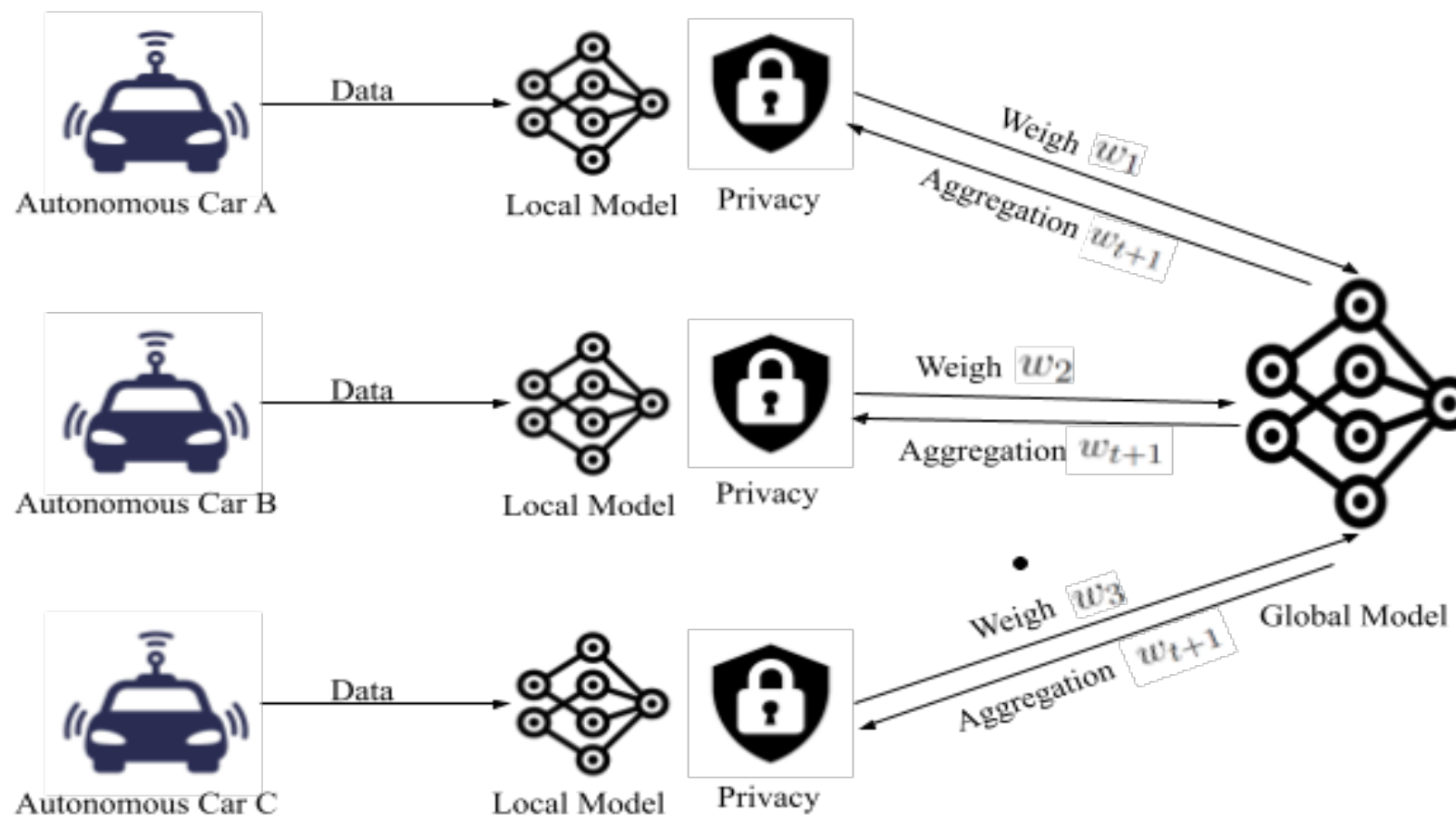
Anonymized dataset with 2-Anonymity

Homomorphic Encryption enables the processing of encrypted data without decryption, ensuring that sensitive information remains secure and accessible only in encrypted form to service providers [3].



An Example of Concept Illustration of Homomorphic Encryption

Federated Learning is a machine learning approach where an algorithm is trained across multiple decentralized devices using local data samples without data exchange, coordinated by a central server to enhance privacy and efficiency [4].



Federated Learning for Secure Model Enhancement

Comparison of Privacy Techniques

Technique	Advantage	Disadvantage
Traditional Data Anonymization	- Simple implementation - Quick execution	- Reduced data utility - Limited flexibility
K-Anonymity	- Effective against re-identification - Ease of implementation	- Data quality reduction - Dependence on the value of 'k'
Homomorphic Encryption	- Secure data processing - Wide applicability	- Limited scalability - Complex implementation
Federated Learning	- Data remains local - Scalable across organizations	- Data heterogeneity - Computation costs

Open Research Problems

- Federated Learning:** Development of techniques for addressing and mitigating bias in federated learning models. & Balancing computational demands with privacy on edge devices.
- Homomorphic Encryption:** Development of lighter versions for resource-constrained edge devices.
- AI-Driven Anonymization:** Development AI methods like transformers, generative AI for better data privacy in edge computing.
- Security, Privacy, and Trust:** Development of mechanisms that tradeoff security, privacy, and trust.
- Legal and Ethical Regulations:** Critical to address for compliance and user protection in edge computing.

References

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